

Question	Marking Guidance	Mark	Comments
09.1	1. <u>Antigen</u> (at T and substrate); 2. Enzyme-substrate complex (produces a line/colour change) OR Enzyme (binds) with substrate (produces line/colour change);	2 (2 x AO2)	1. Reject antigen in blood 2. Accept colourless dye for substrate 2. Accept ES complex in this instance
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09.2	Blood/sample has moved/diffused (above T in the test);	1 (AO3)	Accept Blood/sample and anti-human antibody have moved (in the test)
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09.3	1. (Cancer/fused cells) divide/replicate rapidly/uncontrollably; 2. B cells produce (monoclonal) antibody;	2 (2 x AO2)	1. Accept mitosis OR reproduce for divide 1. Accept hybridoma for 'fused cell' 2. Accept 'plasma cells' OR 'memory cells' for 'B cells' 2. Accept secrete OR make for produce
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09.4	Harmful but not killed OR Harmful but only used once OR Harmful but stops human suffering OR Harmful but produces (useful) medicine/drugs OR Not harmed but injected (with a substance);	1 (AO3)	Must have idea 'for' and idea 'against' Accept stressed OR exploited OR mistreated OR abused for harmed Accept illness OR infection OR death for suffering

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09.5	1. Better than current at detecting early but not as good as lab-based; 2. New/current test better than lab-based from 5 days OR New test as good as current from 5 days; 3. New test and lab-based better total (of) positives than current OR New test total (of) positives not (quite) as good as lab-based; 4. New test (likely to be) quicker; 5. New test (likely to be) cheaper; 6. Limited/inaccessible labs OR Limited training of people to use labs;	3 max (3 x AO3)	Max 2 from 4, 5 and 6 Accept ELISA test for new test 1, 2 and 3 Accept correct comparative figures from Table 3 OR calculations (some examples in table below) 1. and 2. Accept higher proportion for better 1. Accept reference to day(s) in range of 1 to 4 days for early 1. Accept identifies OR finds OR spots for detects 1. and 2. Accept 'more accurate' OR 'identifies more' OR 'finds more' OR 'detects more' OR 'is more successful' OR 'is more effective' for "better than" and the converse of these statements for "not as good as" 2. Accept after 4 days 4. Accept 'more efficient' for quicker

Time after onset of symptoms / days	Number of positive results / % (combined with earlier result(s))		
	Lab-based test	Current test	New test
1–2	71	0	43
3–4	74 (73)	16 (12)	63 (58)
5–7	44 (66)	78 (29)	78 (63)

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09.6	1. Increases water potential of blood/capillary OR Decreases water potential of tissue fluid; 2. (So) less <u>water</u> returns to blood/capillaries (by osmosis) OR (So) more <u>water</u> leaves blood/capillaries (by osmosis);	2 (2 x AO2)	1. Accept Ψ for water potential 1 Ignore WP 1. Accept reduces water potential gradient 2. Ignore tissue fluid 2. Accept 'no' for less
